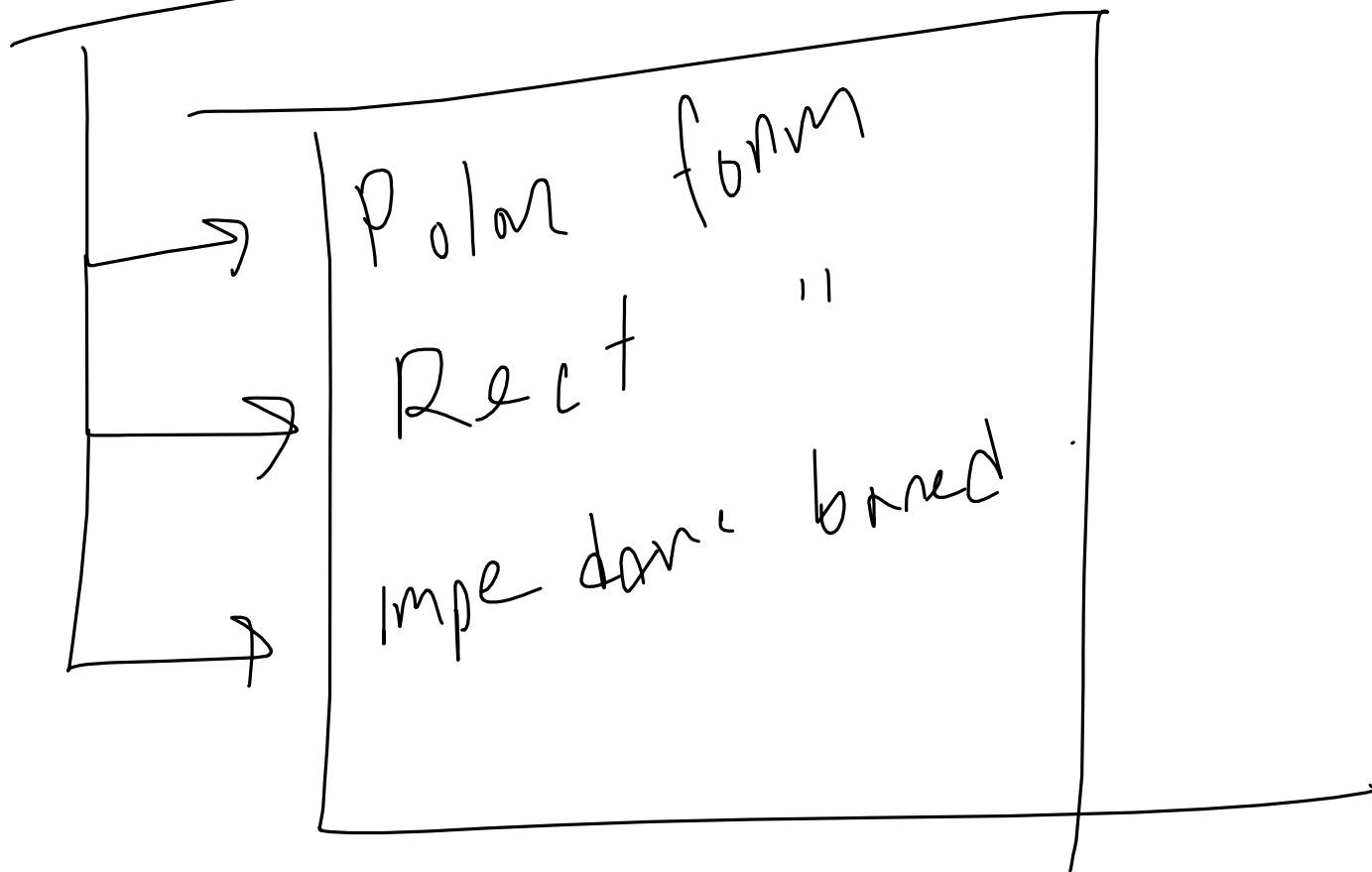


AC Power formula



Power Calculation (polar)

Let
real power

$$V = 50 \angle 30^\circ, \quad I = 100 \angle 50^\circ \checkmark$$

$$\begin{aligned} P &= |V| |I| \times \cos(\theta_v - \theta_i) \\ &= 50 \times 100 \times \cos(30^\circ - 50^\circ) \\ &= \end{aligned}$$

Reactive
Power, Q

$$Q = |V| |I| \sin(\theta_v - \theta_i)$$

$$\text{Complex power} = 58 \times 100 \times \sin(30^\circ - 50^\circ)$$

$$S = P + jQ$$

$$\text{Apparent power} \quad S = \sqrt{P^2 + Q^2}$$

$$\text{Let, } V = 30 + j50 \text{ V, } I = 60 + j70 \text{ A}$$

Rectangular

$$P = \operatorname{Re}\{V I^*\}$$

$$\begin{aligned} V \times I^* &= (30 + j50)(60 - j70) \\ &= 5300 + j900 \end{aligned}$$

$$P = \operatorname{Re}\{VI^*\}$$

$$= 5300 \text{ W}$$

$$Q = \operatorname{Im}\{VI^*\}$$

$$= 900 \text{ VAR}$$

$$S = \sqrt{P^2 + Q^2}$$

$$\underline{S} = P + jQ$$

Let

$$Z = 60 + j80$$

$$Z = R + jX$$

$$P = |I|^2 R$$
$$= (60.42)^2 \times 60$$

$$I = 25 + j55$$

Impedance

$$|I| = 60.42$$

$$Q = |I|^2 \times 2 \times 80$$

$$= (60.4^2) \times 80$$

$$S = \sqrt{P^2 + Q^2}$$

$$\bar{S} = P + jQ$$

Power Factor

$$pf = \cos(\theta_v - \theta_i)$$

$$= \cos \theta_z$$

$\frac{P}{S}$

→

$$V = I Z$$

$$\bar{Z} = \frac{\bar{V}}{\bar{I}}$$

$$= \frac{V \angle \theta_v}{I \angle \theta_i}$$

$$= \frac{V}{I} \angle (\theta_v - \theta_i)$$

How much of apparent power in

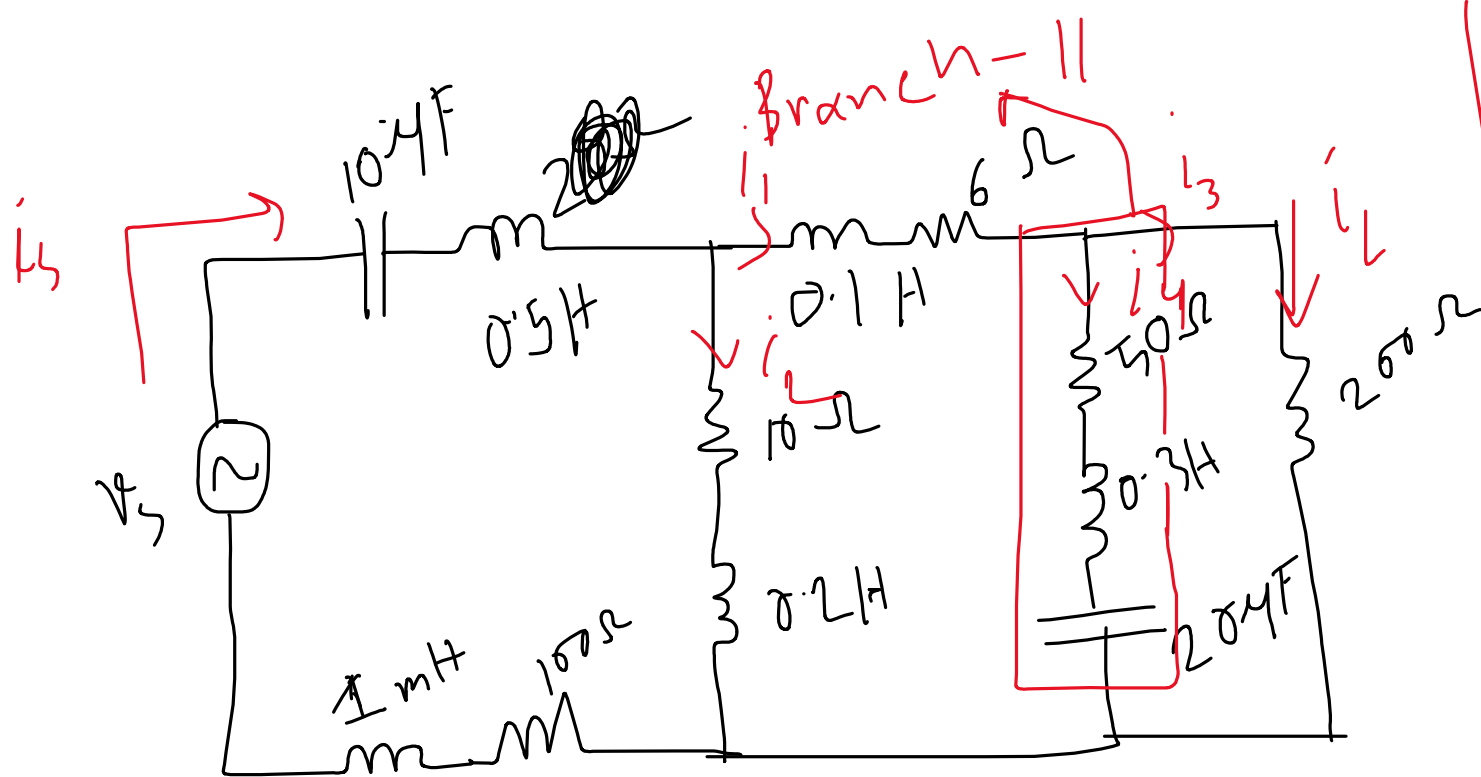
real power $\rightarrow$ AC

$$\frac{P}{S} = \frac{6}{10} = 0.6$$

$\rightarrow 0.9$

0.95

$S = \sqrt{P^2 + Q^2}$



$$V_s = 300 \sin(500\pi t + 60^\circ)$$

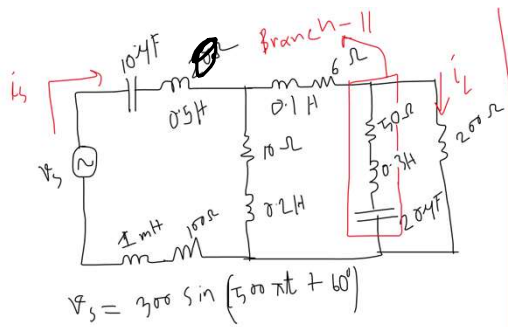
Find $\rightarrow$

~~(i)~~ $i_s = ?$, $P_f = ?$

(ii) $i_L = ?$

(iii) P_s , Q_s ,
 S_s , $\overline{S_s}$

(iv) find P , Q ,
 S , $\overline{S}$ for
 Branch-II



Find →
 (i) $i_s = ?$
 (ii) $i_L = ?$
 (iii) $P_s, Q_s,$
 $S_s, \bar{S}_s$
 (iv) find $P, Q,$
 $S, \bar{S}$ for
 Branch-II

$$V_s = 300 \angle 60^\circ$$

$$10 \mu F \rightarrow$$

$$\frac{1}{500 \times 10 \mu}$$

$$= 200 \Omega$$

$$20 \mu F \rightarrow$$

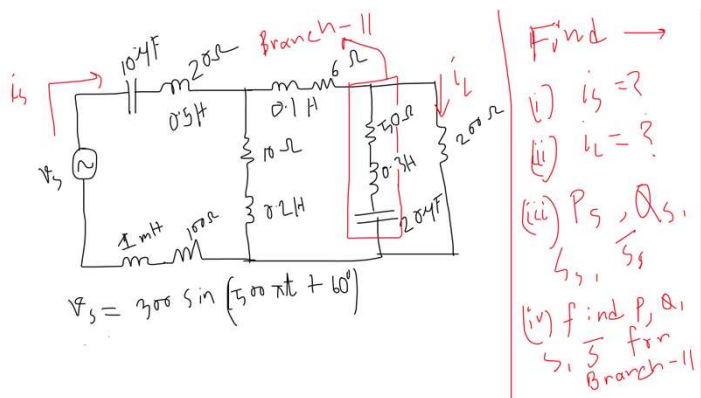
$$\frac{1}{500 \times 20 \mu}$$

$$= 100 \Omega$$

$$0.5 H$$

$$0.5 \times 500$$

$$= 250 \Omega$$

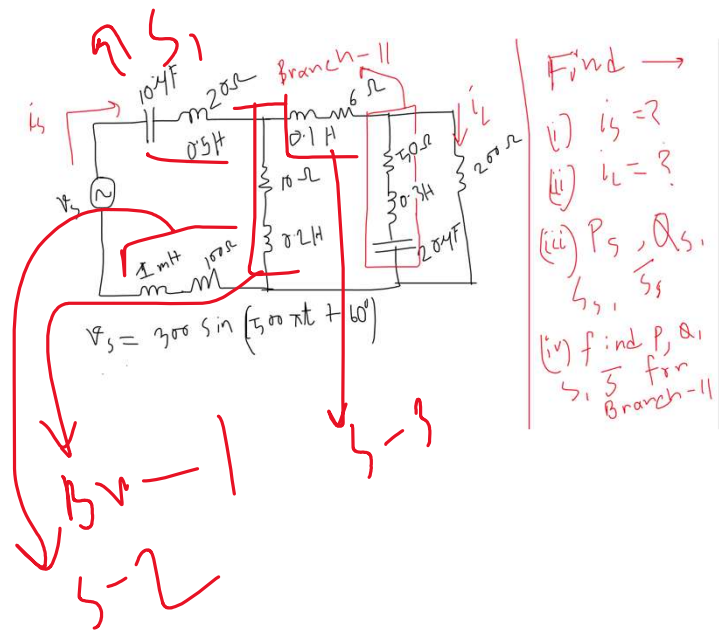


$$1 \text{ mH} \rightarrow 500 \times 1 \text{ m} = 0.5 \Omega$$

$$0.2 \text{ H} \rightarrow 500 \times 0.2 = 100 \Omega$$

$$0.1 \text{ H} \rightarrow 500 \times 0.1 = 50 \Omega$$

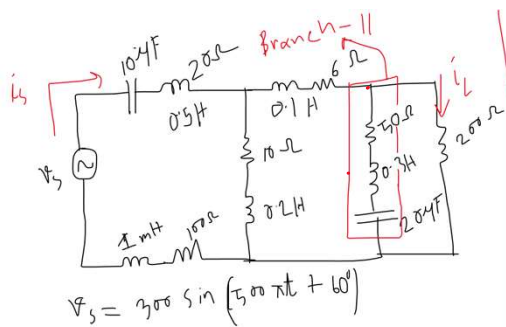
$$0.3 \text{ H} \rightarrow 500 \times 0.3 = 150 \Omega$$



$$I_s \rightarrow I_s$$

$$I_s = \frac{V_s}{Z_{eq}}$$

$$\begin{aligned} Z_{Br-2} &= 50 + j150 - j100 \\ &= 50 + 50j \end{aligned}$$



Find $\rightarrow$

- $i_s = ?$
- $i_L = ?$
- P_s, Q_s, S_s
- find P, Q, S for Branch-II

$$Z_{s3} = 6 + j50$$

$$Z_{Br-1} = 10 + j100$$

$$Z_{s1} = -j200 + j250$$

$$= 50j$$

$$Z_{s2} = 100 + j0.5$$

$$z_{eq} = \left[\begin{array}{c|c} 200 & z_{Br-2} + z_{s3} \\ \hline & z_{Br-1} \end{array} \right]$$

$$52.15 + j80.77$$

$$+ z_{s1} + z_{s2}$$

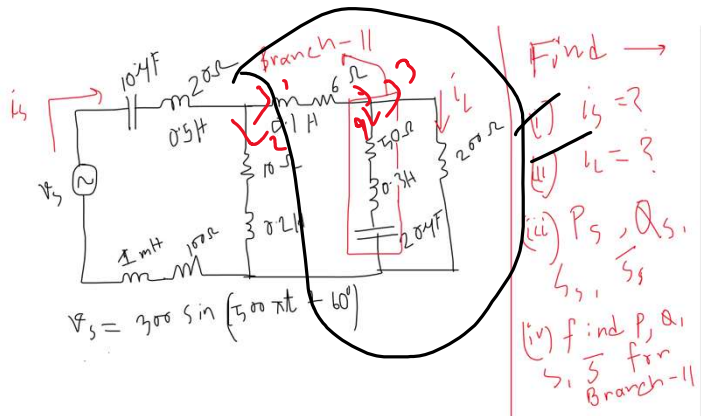
$$16.95 + j47.62$$

$$116.95 + j98.12$$

$$I_s = \frac{V_s}{Z_{eq}} \rightarrow 300 \angle 60^\circ$$

$$= 152 \angle 40^\circ$$

$$PF = \frac{\cos(\theta_v - \theta_i)}{\cos(60^\circ - 40^\circ)} = 0.94$$



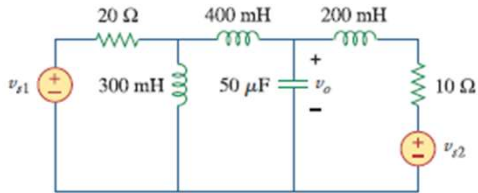
$$I_1 = \frac{(10 + j100)}{(10 + j100) + 52.15 + j80.7}$$

$$= 0.52 \angle 13.2^\circ$$

$$I_3 = I_L = \frac{50 + j50}{50 + j50 + 200} \times I_1$$

$$= 0.14 \angle 47^\circ$$

10.30 Use mesh analysis to find v_o in the circuit of Fig. 10.78. Let $v_{s1} = 120 \cos(100t + 90^\circ)$ V, $v_{s2} = 80 \cos 100t$ V.



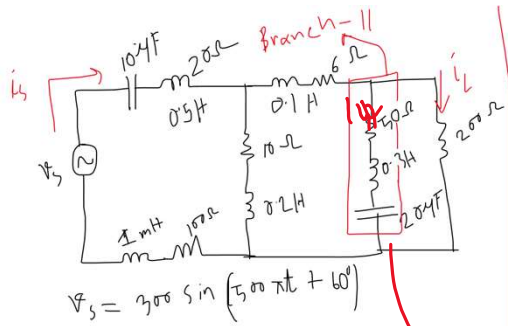
$$P_s = N_s |I_s| \cos(\theta_v - \theta_i)$$

$$= 300 \times 152 \times 0.94$$

$$Q_s = |V_s| |I_s| \sin(\theta_v - \theta_i)$$

$$\overline{S}_s = P_s + j \theta_s$$

$$S_s = \sqrt{P_s^2 + \theta_s^2}$$



Find $\rightarrow$

- (i) $i_s = ?$
- (ii) $i_L = ?$
- (iii) P_s, Q_s
 $S_s, \bar{S}_s$
- (iv) find P, Q
 $S, \bar{S}$ for
 Branch-11

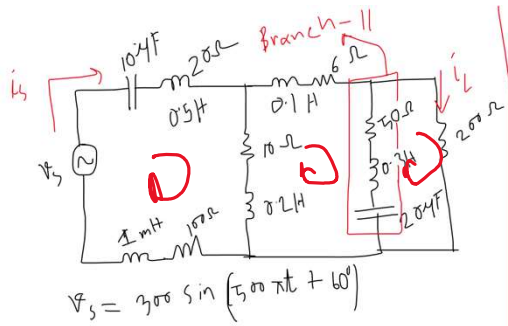
14

Br-2

mesh analysis

$$i_4 = \frac{200}{200 + 50 + j50} \times I_1 \rightarrow 0.52 \angle 13.26^\circ$$

$$= 0.78 \angle -11.3^\circ$$



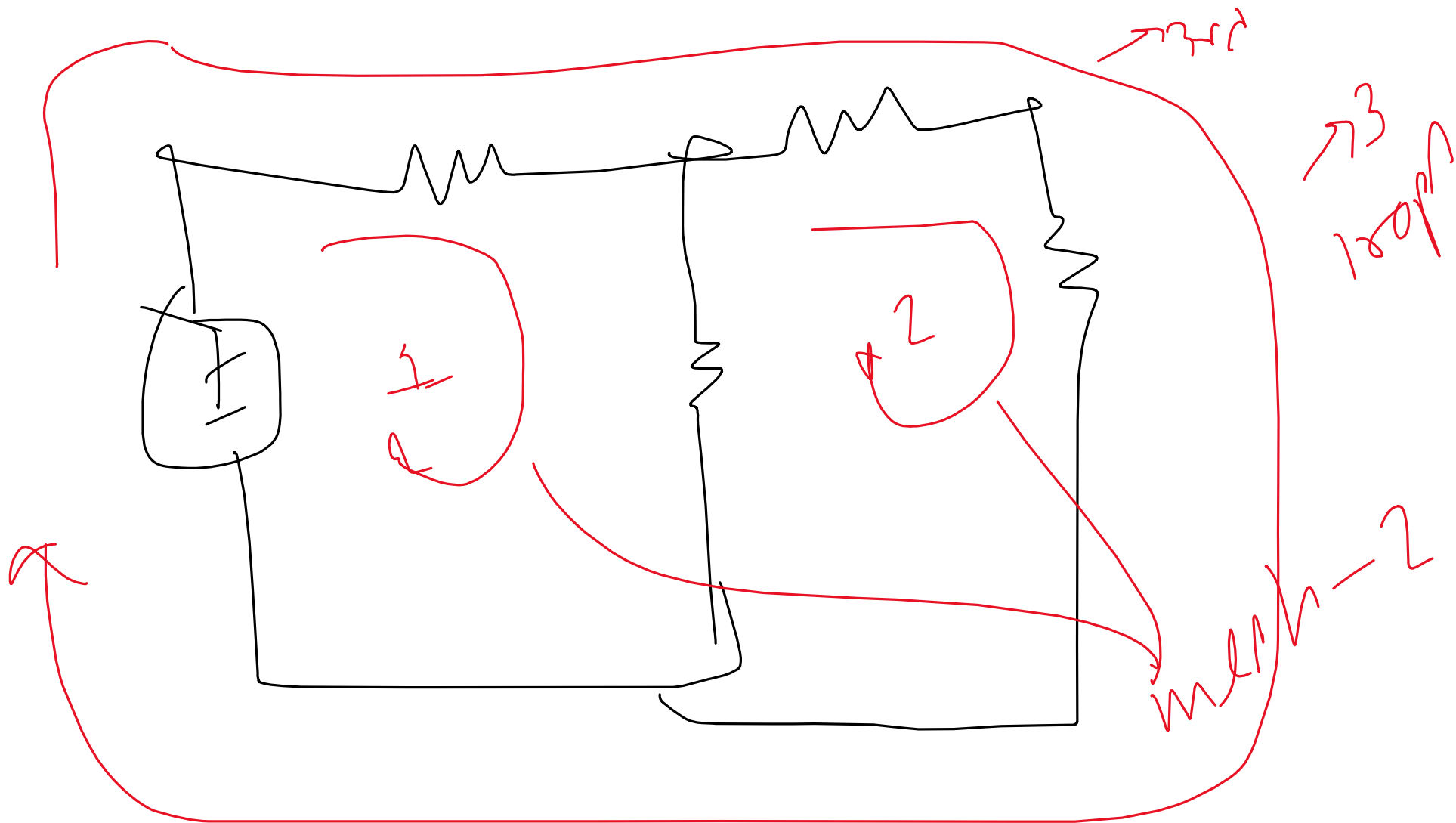
Find $\rightarrow$

- $i_s = ?$
- $i_L = ?$
- P_s, Q_s, S_s
- find P, Q, S for Branch-II

Mesh-1

Mesh Analysis

- ① find meshes in ckt
- ② Apply KVL to meshes
- ③ Soln $\rightarrow$ KVL eqn.



memh = independent loop

↳ loop that has no other loop within.